

MODULE4

RobotFramework

Module	RobotFramework
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EXPERIMENT 4.1 – Introduction to Robot Framework

1. Problem Statement

To understand the basic syntax of Robot Framework and create a simple web automation test using SeleniumLibrary.

2. Objective

To setup Robot Framework, use basic keywords, open a webpage, enter text and verify an element.

3. Tools, Software & Concepts Used

- Python
- Robot Framework
- SeleniumLibrary
- VSCode/PyCharm
- Web browser

Concepts: Robot Framework syntax, test cases, keywords, SeleniumLibrary, Input Text, Press Keys and Page Should Contain

4. Source Code

```
***Settings***
```

```
Library    SeleniumLibrary
```

```
***Test Cases***
```

```
Open Website And Search
```

```
    Open Browser    https://www.bing.com
```

```
    chrome Maximize Browser Window
```

```
    Input Text    name=q
```

```
    Robot Framework Press Keys
```

```
    name=q    ENTER
```

```
    Wait Until Page Contains Robot Framework
```

```
    10s Page Should Contain Robot Framework
```

```
    Sleep    2
```

```
    Close Browse
```

```
r
```

5. Output

5.1 TERMINALOUTPUT

```
PS C:\Users\debas\PycharmProjects\PythonProject\mod4> robot exp4.1.robot
=====
Exp4.1
=====
Open Website And Search | PASS |
-----
Exp4.1 | PASS |
1 test, 1 passed, 0 failed
=====
Output: C:\Users\debas\PycharmProjects\PythonProject\mod4\output.xml
Log: C:\Users\debas\PycharmProjects\PythonProject\mod4\log.html
Report: C:\Users\debas\PycharmProjects\PythonProject\mod4\report.html

PS C:\Users\debas\PycharmProjects\PythonProject\mod4>
```

Fig4.1.1:RobotFrameworkshouldshowtheexecutedtestcaseanditsPASS/FAIL result.

5.2 SELINIUMBROWSEROUTPUT

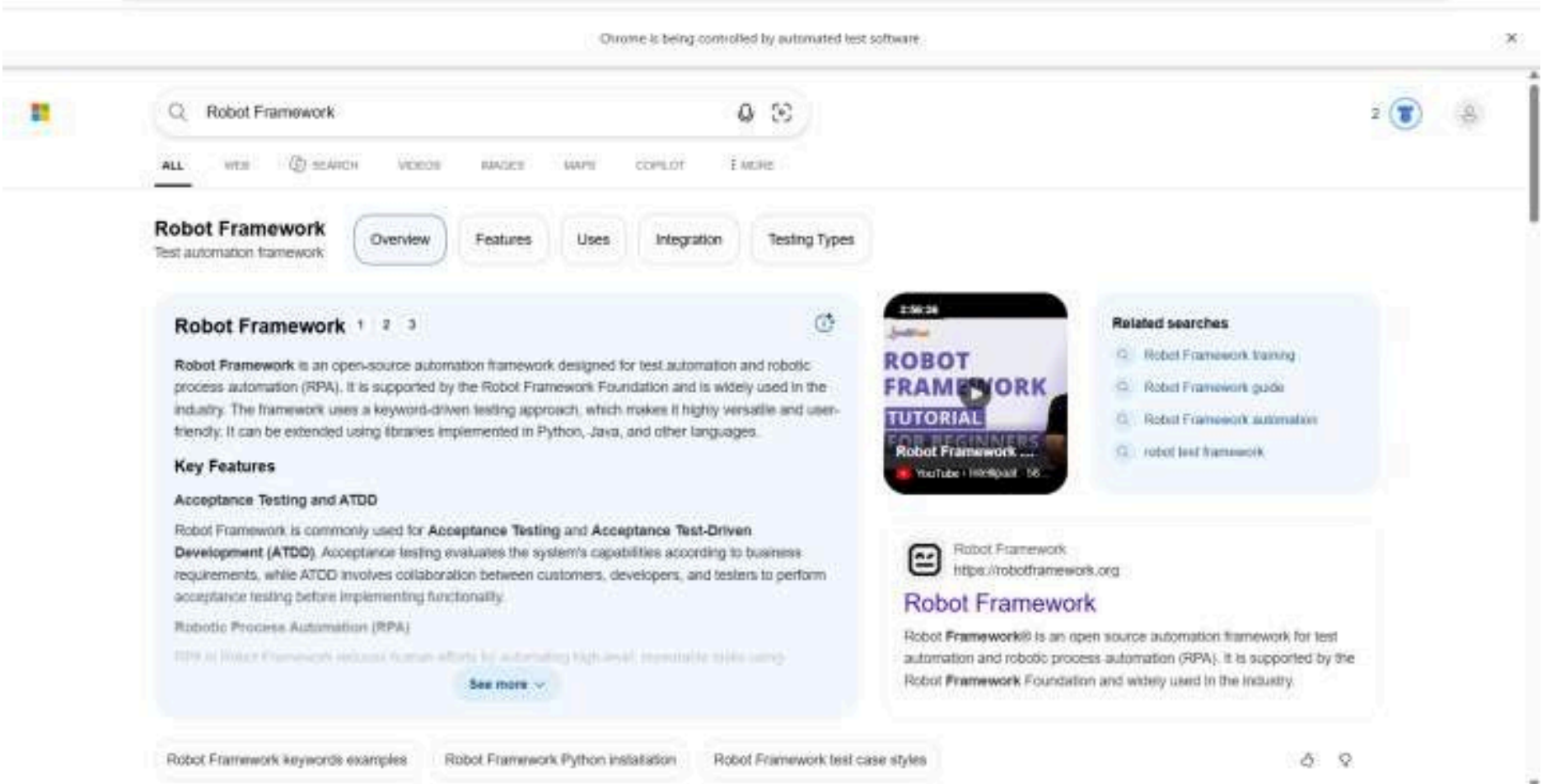


Fig4.1.2:Theautomatedtaskcompletedsuccessfully.

5.3 log.htmlGenerated

Exp4.1 Log

Generated
20260919 23:41:25 UTC+05:30
32 seconds ago

Statistics

Total Statistics	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
All Tests	1	1	0	0	00:00:07	
Statistics by Tag	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
No Tags						
Statistics by Suite	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
Exp4.1	1	1	0	0	00:00:07	

Execution Log

- SUITE Exp4.1	00:00:07.347
Full Name: Exp4.1 Source: C:\Users\debas\PycharmProjects\PythonProject\mod4\exp4.1.robot Start / End / Elapsed: 20260919 23:41:18.831 / 20260919 23:41:25.878 / 00:00:07.247 Status: 1 test total, 1 passed, 0 failed, 0 skipped	
+ TEST Open Website And Search	00:00:06.992

Fig 4.1.3:screenshotshowingExperiment4_1loggeneration

5.4report.htmlGenerated



Fig 4.1.4:ScreenshotShowingExperiment4_1reportgeneration

6. Result,Observation&Conclusion

Result:RobotFrameworkexecutedthetestcasesuccessfully.

Observation: The browser was opened, text was entered using a keyword and the expected text was checked on the page. Robot Framework also generated output.xml, log.html, and report.html files automatically.

Conclusion: This experiment demonstrated the basic use of Robot Framework with SeleniumLibrary for web automation. The test case was written in a .robot file using Robot Framework syntax. The successful terminaloutputconfirmedthat the complete test case executed without errors. The Selenium browser also showed that the required web actions were performed correctly. Thus, Robot Framework can be used to create simple and readable web automation test cases.

EXPERIMENT4.2–ScriptRunningOptions

1. ProblemStatement

To understand basic ways of running Robot Framework test cases from the command line and the result files generated after execution.

2. Objective

To run a Robot Framework test from the terminal and understand setup, teardown and generated reports.

3. Tools,Software&ConceptsUsed

- Python
- Robot Framework
- SeleniumLibrary
- PyCharmTerminal/CommandPrompt

Concepts: robot command, test setup and teardown, output.xml, log.html, report.html and running test suites

4. SourceCode

```
***Settings***
Library    SeleniumLibrary

***Test Cases***
CheckGoogle
Page
    Open Browser    https://www.google.com
    chrome Maximize Browser Window
    Title Should Be
    GooglePage Should Contain
        Google Sleep    2
    CloseBrowser
```

5. Output

5.1 TERMINALOUTPUT

```
PS C:\Users\debas\PycharmProjects\PythonProject\mod4> robot exp4.2.robot
=====
Exp4.2
=====
Check Google Page | PASS |
-----
Exp4.2 | PASS |
1 test, 1 passed, 0 failed
=====
Output: C:\Users\debas\PycharmProjects\PythonProject\mod4\output.xml
Log:    C:\Users\debas\PycharmProjects\PythonProject\mod4\log.html
Report: C:\Users\debas\PycharmProjects\PythonProject\mod4\report.html

PS C:\Users\debas\PycharmProjects\PythonProject\mod4>
```

Fig4.2.1:RobotFrameworkshowingtheexecutedtestcaseanditsPASS/FAIL result.

5.2SELINIUMBROWSEROUTPUT

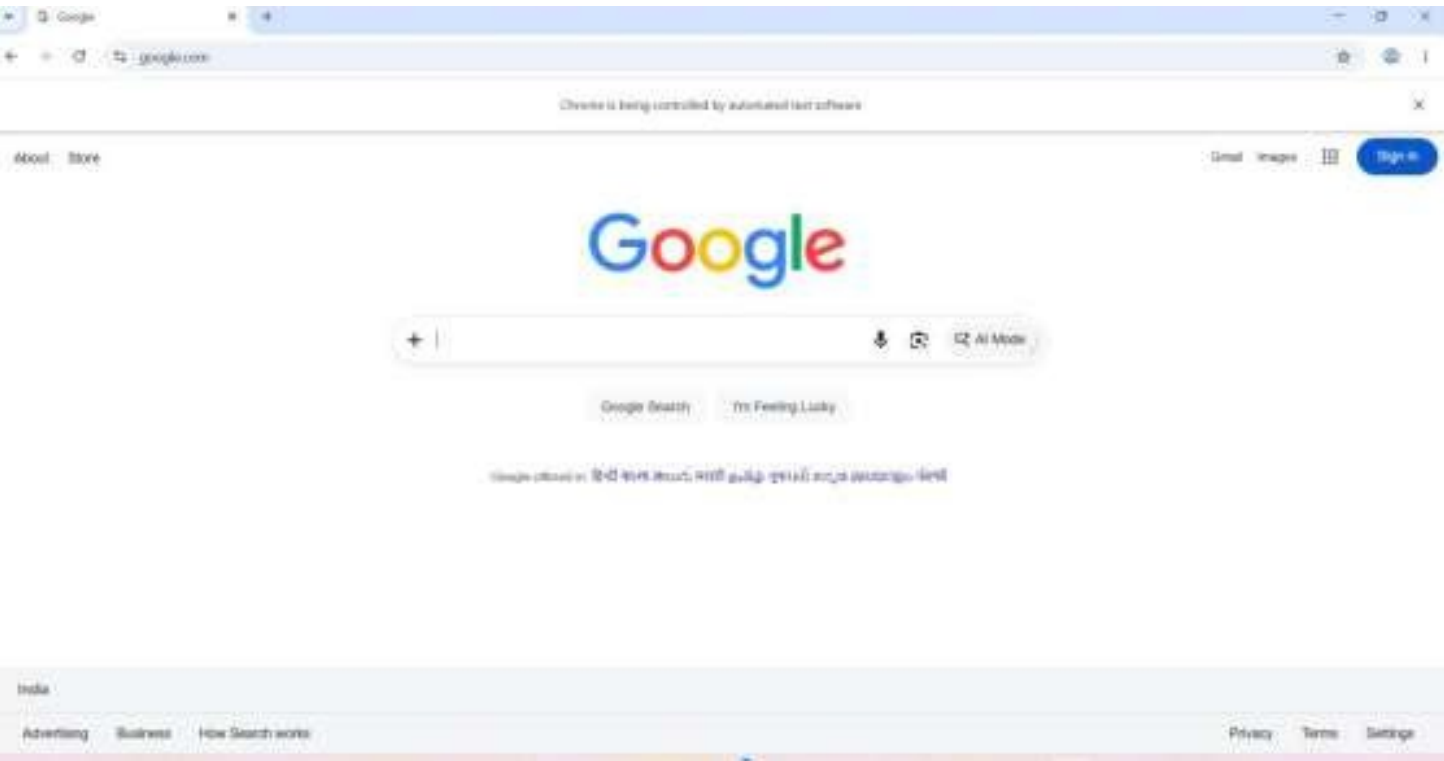


Fig4.2.2:Theautomatedtaskcompletedsuccessfully.

5.3log.htmlGenerated

REPORT

Experiment 4 2 Log

Generated: 20250919 02:52:13 UTC+05:30
4 minutes 8 seconds ago

Statistics

Total Statistics	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
All Tests	1	1	0	0	00:00:07	

Statistics by Tag	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
No Tags						

Statistics by Suite	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
Experiment 4 2	1	1	0	0	00:00:07	

Execution Log

- Suite: Experiment 4 2

Full Name: Experiment 4 2

Source: C:\Users\Shamir\OneDrive\Documents\1\atReport\Module4\Experiment_4_2.robot

Start / End / Elapsed: 20250919 02:52:06.044 / 20250919 02:52:13.233 / 00:00:07.189

Status: 1 test total, 1 passed, 0 failed, 0 skipped

+ Test: Check Google Page

00:00:06.016

Fig 4.2.3:screenshotshowingExperiment4_2loggeneration

5.4report.htmlGenerated



Fig 4.2.4:ScreenshotShowingExperiment4_2reportgeneration

6. Result,Observation&Conclusion

Result:

TheRobotFrameworkscriptwas executedsuccessfullyusing SeleniumLibrary. The Google webpage was openedin Chrome,the page titleandpagecontentwere checked,and thebrowser was closed successfully. The terminal showed **1 test passed and 0 failed**.

Observation:

The test case was executed using Robot Framework keywords in a simple and readable format. The Open Browser keyword opened the webpage, Title Should Be and Page Should Contain were used for verification, and Close Browser closed the browser after execution. Robot Framework also generated output.xml, log.html, and report.html files automatically.

Conclusion:

This experiment demonstrated different options for running a Robot Framework test script from the terminal.Thesuccessfulterminaloutput confirmedthat thetestwas executedwithout errors. The generated XML and HTML filesprovidedadditionalexecutioninformationandtestresults.The experiment helped in understanding how Robot Framework runs test cases and records their execution results. Thus, the basic script-running process of Robot Framework was successfully demonstrated.

EXPERIMENT 4.3 – Make the Script More Readable

1. Problem Statement

To improve Robot Framework scripts by using user-defined keywords, resource files and setup or teardown.

2. Objective

To break a test into meaningful reusable keywords and keep common actions in a resource file.

3. Tools, Software & Concepts Used

- Python
- Robot Framework
- Selenium Library
- Resource files
- VS Code/PyCharm

Concepts: Procedural vs Gherkin style, user-defined keywords, resource files, setup and teardown and reusable actions

4. Source Code

```
#resources/common.robot
***Settings***
Library SeleniumLibrary

*** Keywords ***
Open Test Website
    Open
    Browser https://www.saucedemo.com/chromeMaximize Browser Window

Close Test Browser
    Close All Browsers

Login With Username And Password
    [Arguments] ${username} ${password}
    Input Text id=username ${username}
    Input Text id=password ${password}
    Click Button id=login-button

#Experiment_4_3.robot
***Settings***
Resource resources/common.robot
Suite Teardown Close Test Browser

***Test Cases***
Login Using Reusable Keyword
    Open Test Website
    Login With Username And Password
    standard_user secret_sauce Wait Until Page Contains Products 10 seconds
    Sleep 2
```

5. Output

5.1 TERMINALOUTPUT

```
PS C:\Users\debas\PycharmProjects\PythonProject\mod4> robot exp4.3.robot
=====
Exp4.3
=====
Login Using Reusable Keyword | PASS |
-----
Exp4.3 | PASS |
1 test, 1 passed, 0 failed
=====
Output: C:\Users\debas\PycharmProjects\PythonProject\mod4\output.xml
Log: C:\Users\debas\PycharmProjects\PythonProject\mod4\log.html
Report: C:\Users\debas\PycharmProjects\PythonProject\mod4\report.html

PS C:\Users\debas\PycharmProjects\PythonProject\mod4>
```

Fig4.3.1:RobotFrameworkshowingtheexecutedtestcaseanditsPASS/FAIL result.

5.2ROBOTFRAMEWORK/BROWSEROUTPUT

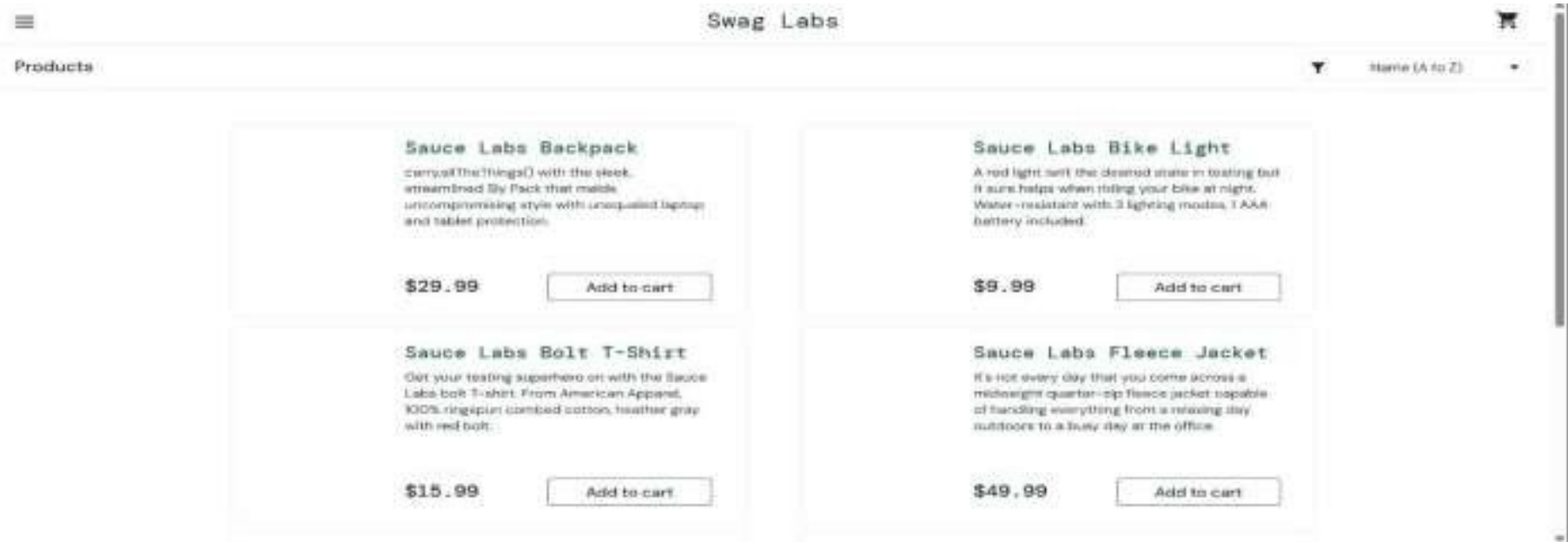


Fig4.3.2:Theautomatedtaskcompletedsuccessfully.

5.3log.htmlGenerated



Fig 4.2.3:screenshotshowingExperiment4_3loggeneration

5.4report.htmlGenerated



Fig 4.2.4:ScreenshotShowingExperiment4_2reportgeneration

5.5FilesGeneratedAfterExecution

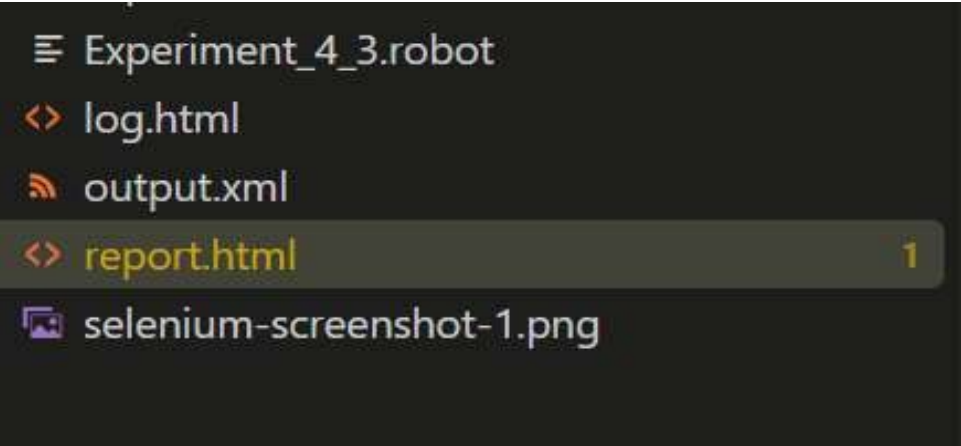


Fig4.2.5:Screenshotshowingoutput.xml,selenium-screenshot-1.png,log.htmlandreport.htmlgenerated

6. Result,Observation&Conclusion

Result:

The Robot Framework test was executed successfully using a reusable resource file and SeleniumLibrary. The browserwas opened,theSauceDemo loginwas performed using a reusable keyword, and the Products page was verified successfully. The terminal showed **1 test passed and 0 failed**.

Observation:

The common Selenium operations were placed in common.robot and reused in Experiment_4_3.robot. Custom keywords such as Open Test Website, Login With Username And Password, and Close Test Browser made the test case shorter and easier to understand. The Wait Until Page Contains keyword was also used to wait for the Products page before verification.

Conclusion:

This experiment demonstrated how Robot Framework scripts can be made more readable and reusable using resource files and custom keywords. Instead of writing all Selenium commands directly in the test case, common operations were separated into `common.robot`. The successful terminal output confirmed that the reusable keywords worked correctly. The experiment also showed how reusable components can make automation scripts easier to maintain and understand. Thus, the objective of making the Robot Framework script more readable was successfully achieved.